# Intelligent Document Classification System

SparkIIT AI Internship Project 1 | NLP, Feature Extraction, Classification, Model Evaluation

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## Objective

To build an AI-based system that automatically classifies text documents into topic categories, and to evaluate it rigorously using Accuracy, Precision, Recall and F1-Score. The project covers the full machine learning pipeline: data collection, text cleaning, feature extraction, model training, optimization and error analysis.

## Dataset

The **20 Newsgroups** corpus (Usenet discussion posts, built into scikit-learn) was used with 6 categories: comp.graphics, rec.autos, rec.sport.hockey, sci.med, sci.space and talk.politics.guns. The data has 5,668 documents and is roughly balanced (about 880 to 970 documents per class). Headers, footers and quoted replies were removed so that the model learns from message content rather than metadata. Posts are short (median 85 words, mean 185 words). Empty documents were dropped, and a stratified 80/20 split left **1,131 documents for testing**.

## Key Work Done

- Explored the data: class balance and document length distribution.
- Cleaned the text: lowercasing, letters-only tokenization and stop-word removal.
- Extracted features with **TF-IDF** (unigrams and bigrams, 20,000 features, min_df = 2), fitted on training data only to prevent data leakage.
- Trained and compared three classifiers: Multinomial Naive Bayes, Logistic Regression and Linear SVM.
- Evaluated with Accuracy, weighted Precision, Recall and F1-Score, per-class metrics and a confusion matrix.
- Optimized the best model with GridSearchCV (3-fold cross-validation over C = 0.05, 0.1, 0.5, 1, 2, 5); best C = 0.5.
- Compared against a modern baseline: sentence embeddings (all-MiniLM-L6-v2) with Logistic Regression.
- Performed error analysis: most confused category pairs and the most influential words per category.
- Tested the final model on unseen, hand-written sentences.

## Results

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Naive Bayes | 0.9045 | 0.9058 | 0.9045 | 0.9044 |
| Logistic Regression | 0.8974 | 0.8977 | 0.8974 | 0.8974 |
| Linear SVM | 0.9089 | 0.9090 | 0.9089 | 0.9088 |
| Linear SVM (tuned, C=0.5) | 0.9107 | 0.9108 | 0.9107 | 0.9105 |
| MiniLM embeddings + Logistic Regression | 0.8930 | n/a | 0.8930 | 0.8930 |

Table 1. Test-set results (precision, recall and F1 are support-weighted averages). Tuned-model precision and recall are computed from its confusion matrix. Precision for the embedding baseline was not recorded.

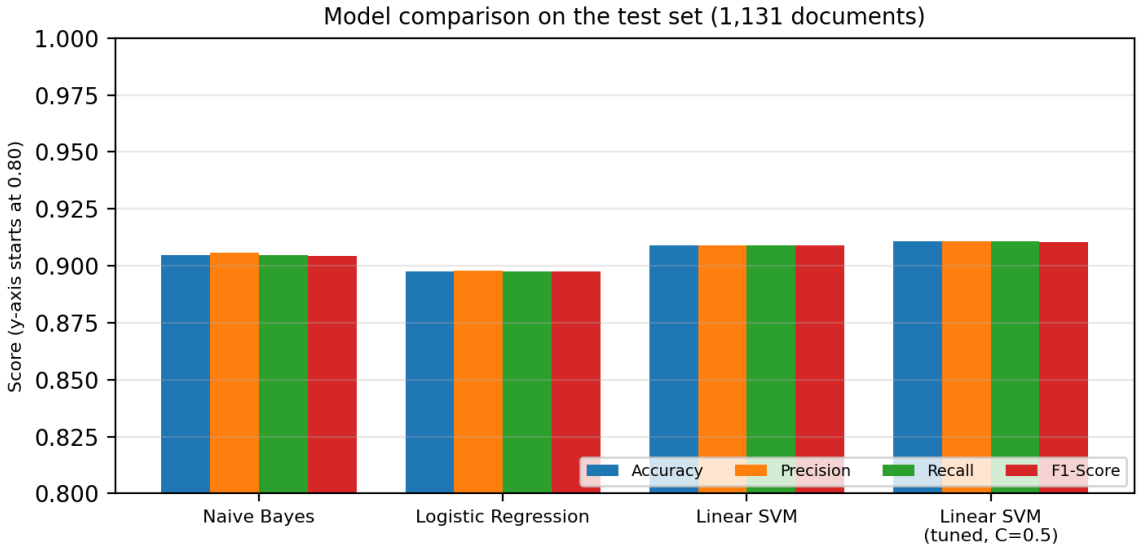



Figure 1. Model comparison.

| Category | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| comp.graphics | 0.905 | 0.953 | 0.928 | 190 |
| rec.autos | 0.904 | 0.909 | 0.907 | 187 |
| rec.sport.hockey | 0.948 | 0.943 | 0.946 | 194 |
| sci.med | 0.893 | 0.911 | 0.902 | 192 |
| sci.space | 0.903 | 0.880 | 0.891 | 191 |
| talk.politics.guns | 0.911 | 0.864 | 0.887 | 177 |

Table 2. Per-category performance of the tuned Linear SVM (computed from the confusion matrix).

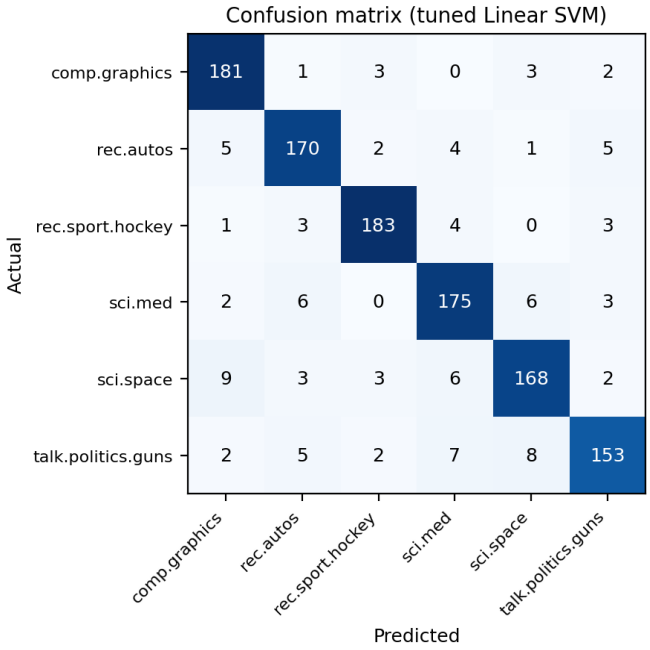



Figure 2. Confusion matrix of the tuned Linear SVM.

## Analysis

- **Model comparison.** All three TF-IDF models finish within about 1.2 points of each other. With 1,131 test documents the standard error of the accuracy is roughly 0.8 points, so the ranking (Linear SVM, then Naive Bayes, then Logistic Regression) is suggestive rather than statistically proven. Tuning raised F1 from 0.9088 to 0.9105, a gain too small to call meaningful. The honest conclusion is that TF-IDF with a linear model works well on this task and the exact choice of linear model matters little.
- **Embedding baseline.** Sentence embeddings (all-MiniLM-L6-v2) with an untuned Logistic Regression reached 0.8930 accuracy, about 1.8 points below the tuned TF-IDF model. Plausible reasons are that these posts are short and driven by topic keywords, which TF-IDF captures directly, and that general-purpose embeddings are built for sentence similarity rather than these specific topics. These explanations were not tested, and the result comes from a single split. Embeddings were also far more expensive: encoding the 4,500 training documents took about 6 minutes 44 seconds on the Colab runtime used, versus seconds for TF-IDF, and they need a large model and library at deployment. This supports using TF-IDF with a Linear SVM here.
- **Strongest and weakest categories.** rec.sport.hockey (F1 0.946) and comp.graphics (F1 0.928) are the easiest, since they have very distinctive vocabulary. talk.politics.guns has the lowest recall (0.864) and sci.space the lowest F1 (0.891).
- **Confusions.** The most frequent errors were sci.space predicted as comp.graphics (9), talk.politics.guns predicted as sci.space (8) and as sci.med (7). A likely explanation is vocabulary overlap: space posts discussing images and files resemble graphics posts, and political posts about government, funding or health borrow words from science categories. This was not verified on the individual documents.
- **What the model learned.** The top words are sensible (gun, firearms, weapons; orbit, shuttle, nasa; puck, nhl, season; car, engine, dealer). Some are corpus-specific quirks: for example 'koresh' (the 1993 Waco standoff), 'roger' (probably a frequently mentioned name) and 'msg' or 'photography' under sci.med. The model therefore partly learns the habits of this 1990s Usenet data and may not transfer directly to modern text.
- **Unseen test.** All 6 hand-written test sentences were classified correctly. With only 6 examples this is a sanity check, not a performance estimate.

## Limitations and Future Work

- A single train/test split was used for the final numbers; k-fold cross-validation would give more reliable estimates.
- TF-IDF ignores word order and meaning. The embedding baseline did not beat it here, but fine-tuning a transformer model end to end, or tuning the classifier on embeddings, might change that and was not tried.
- The corpus is old and covers only 6 topics; testing on modern text, more categories and noisy real-world documents would show how well the system generalizes.
- The model could be wrapped in a small web app so users can paste a document and see the predicted category with a confidence score.

## Outcome

Delivered a complete, leakage-free NLP classification pipeline that reaches about **91% accuracy and 0.91 weighted F1** across six topic categories, with a transparent comparison of three classical models plus an embedding baseline, hyperparameter tuning, per-category evaluation and an honest analysis of where and why it fails. The approach applies directly to real tasks such as routing support tickets, organizing news and triaging documents.